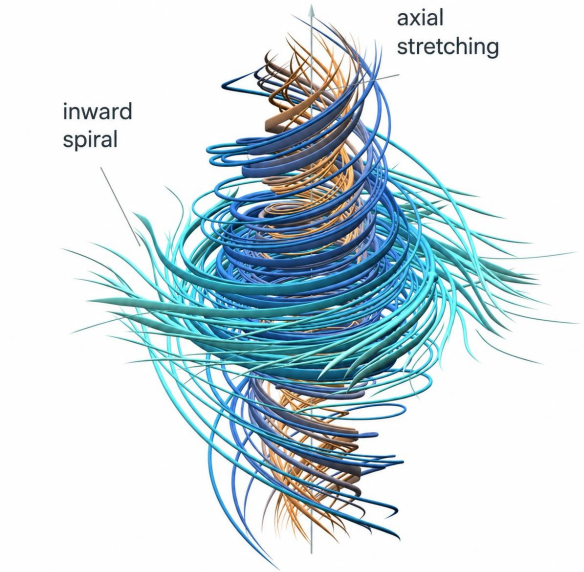


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National Labs Agentic Training

The Apollo Moment for Autonomous Science: Agentic Discovery at Exascale

- Navier-Stokes equations describing fluid motion have remained unsolved as a Millennium Prize Problem since 2000
- OpenAI's internal multi-agent system produced a 166-page proof and a Lean-verified formalization showing smooth fluid flow can develop a singularity in finite time
- Extreme scale:
  - 10,000 concurrent agents
  - 88 hours
  - 4.9 million messages
  - ~300 billion output tokens
  - Estimated millions of dollars



OpenAI, Finite time blowup for Navier-Stokes, https://openai.com/index/navier-stokes-solution/

And this is one of many examples

Science

Claude discovers a novel enzyme system with CRISPR-like repeats

Sep 23, 2026



October 6, 2026 Research Publication

Sharing AI progress in mathematics

View on GitHub ↗

Listen to article | 2:04 Share

We're releasing a broad range of new mathematical results produced by an internal frontier model.

As we look to improve how we share results with the math community, we've been consulting with the independent Advisory Group on Mathematics and Artificial Intelligence at the Institute for Advanced Study to develop best practices, and we have drawn on their advice and

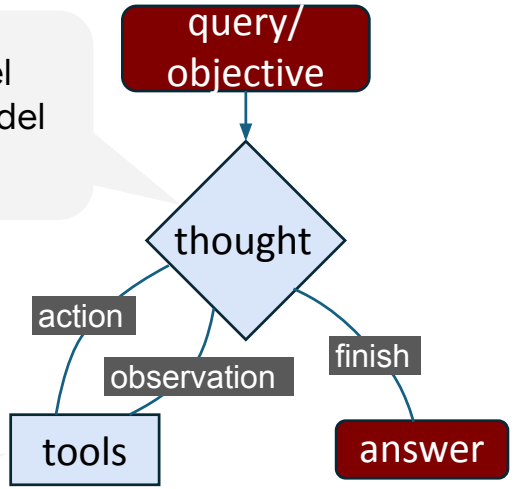
What is an “agent”?

An agent is a persistent, stateful process that acts on behalf of a user or system. An agent may:

- Sense/Observe inputs or events
- Plan (decide on) actions using a policy (rules or LLM)
- Act: Execute tools or call other agents
- Learn: Update state to adapt

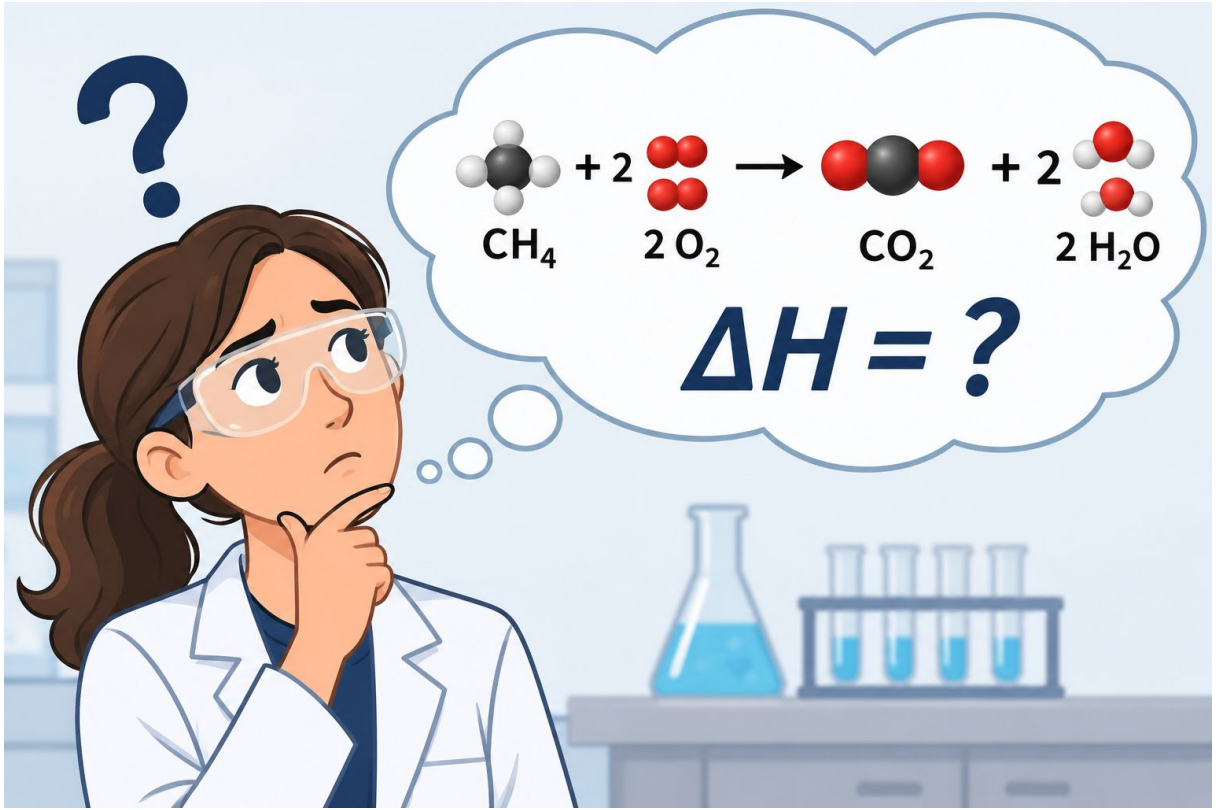
Large Language Model
Specialized Science Model
Vision Transformer

Science Codes
Instruments

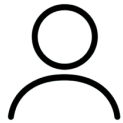


We can think of an agent as a capable colleague that can reason, act, and coordinate autonomously toward an objective

Example: From LLMs to Agents

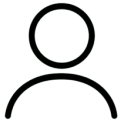


You are given a chemical reaction: 1 (Methane) + 2 (Oxygen) -> 1 (Carbon dioxide) + 2 (Water). Calculate the enthalpy change for this reaction using at 400K



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You are given a chemical reaction: 1 (Methane) + 2 (Oxygen) -> 1 (Carbon dioxide) + 2 (Water). Calculate the enthalpy change for this reaction at 400K



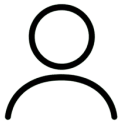


LLM

The reaction enthalpy at 400 K is approximately **-8.3 eV** ✘

* Older hosted LLMs had limited agentic/tool capabilities

You are given a chemical reaction: 1 (Methane) + 2 (Oxygen) -> 1 (Carbon dioxide) + 2 (Water). Calculate the enthalpy change for this reaction using GFN2-xTB at 400K



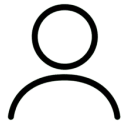


LLM

To actually calculate the enthalpy change **using GFN2-xTB at 400 K**, I would need to run xTB on molecular structures for CH_4, O_2, CO_2, and H_2O and obtain their thermal corrections at 400 K. **I don't have the ability to execute xTB calculations directly.**

* Older hosted LLMs had limited agentic/tool capabilities

You are given a chemical reaction: 1 (Methane) + 2 (Oxygen) -> 1 (Carbon dioxide) + 2 (Water). Calculate the enthalpy change for this reaction using GFN2-xTB at 400K





LLM

The enthalpy change for the reaction at 400 K is approximately -801 kJ/mol

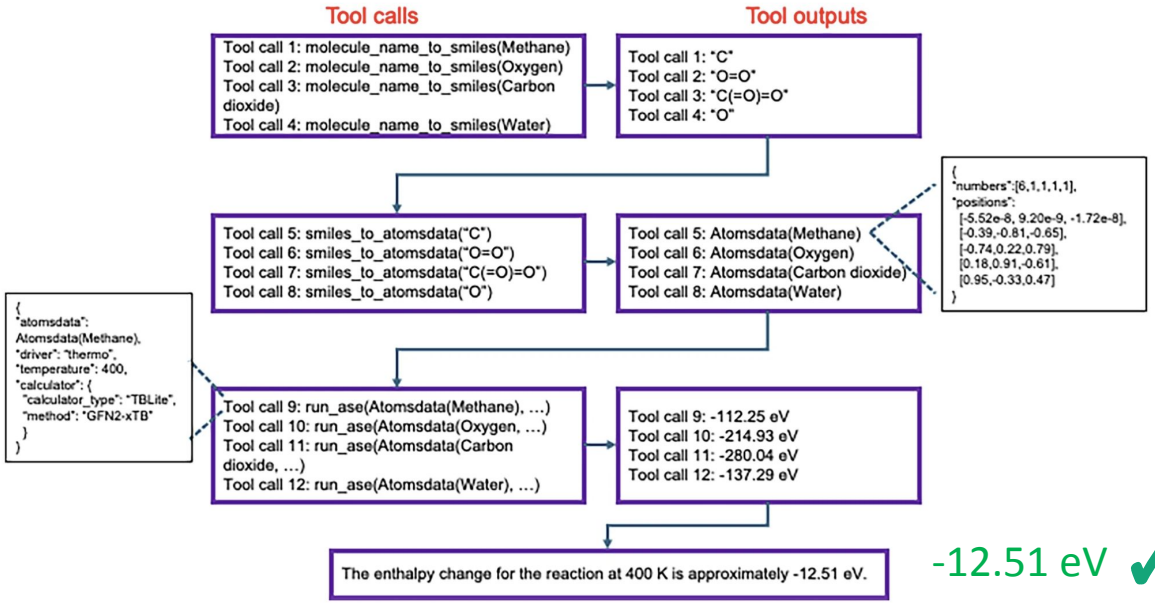
* Newer hosted LLMs have expanded agentic/tool capabilities

You are given a chemical reaction: 1 (Methane) + 2 (Oxygen) -> 1 (Carbon dioxide) + 2 (Water). Calculate the enthalpy change for this reaction using GFN2-xTB at 400K



Murat Keçeli et al.

ChemGraph
agent framework:



-12.51 eV ✓

Toward an agentic scientific method



J. G. Pauloski, K. Chard and I. Foster, "Agentic Discovery: Closing the Loop With Cooperative Agents," in Computer, 2025

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Course overview

1. Explore how agentic systems are built, their core components (models, tools, memory and context management, planning and orchestration loops) and how harnesses and frameworks can be used
2. Build end-to-end agentic pipelines for scientific applications
3. Integrate agentic systems with DOE infrastructure, including HPC systems, scientific instruments, and data resources
4. Scale an agentic system from a single agent to many agents
5. Evaluate agentic systems by defining success criteria and benchmarks, and by measuring whether changes actually improve output
6. Govern agentic systems by identifying risks (bias, safety, and ethics), proposing mitigations and safeguards, and incorporating human oversight